Customer Shopping Behavior Analysis

Data Analytics Project Report

1. Project Overview

This project investigates customer shopping behavior using transactional records from 3,900 purchases spanning multiple product categories. The primary objective is to derive actionable insights into spending patterns, customer segmentation, product preferences, and subscription behavior to support data-driven business decisions.

2. Dataset Summary

The dataset consists of the following key characteristics:

- **Rows:** Total Records
- 3,900 transactions
- **Columns:** Total Features
- 18 attributes

Key feature groups included in the dataset:

  - Customer Demographics — Age, Gender, Location, Subscription Status
  - Purchase Details — Item Purchased, Category, Purchase Amount (USD), Season, Size, Color
  - Shopping Behavior — Discount Applied, Promo Code Used, Previous Purchases, Purchase Frequency, Review Rating, Shipping Type

- **Data Quality Note:** Missing Data
- 37 missing values identified in the Review Rating column, handled during preprocessing.

3. Exploratory Data Analysis (Python)

Data preparation and cleaning were performed in Python using the pandas library. The key steps undertaken are outlined below:

- ### Step 1: Data Loading

  Dataset was imported using pandas and initially inspected via df.info() and df.describe() to understand structure, data types, and distribution.

Dataset Summary Statistics Screenshot:

```
[4]: df.describe()
```

[4]:

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

- ### Step 2: Missing Data Handling

  Null values were identified and the 37 missing Review Rating entries were imputed using the median rating per product category.

Missing Value Check Output:

```
[6]: median_ratings=df.groupby('Category')['Review Rating'].transform('median')
     df['Review Rating']=df['Review Rating'].fillna(median_ratings)
     df.isnull().sum()
```

```
[6]: Customer ID               0
     Age                       0
     Gender                    0
     Item Purchased            0
     Category                  0
     Purchase Amount (USD)     0
     Location                  0
     Size                      0
     Color                     0
     Season                    0
     Review Rating             0
     Subscription Status       0
     Shipping Type             0
     Discount Applied          0
     Promo Code Used           0
     Previous Purchases        0
     Payment Method            0
     Frequency of Purchases    0
     dtype: int64
```

- **Step 3: Column Standardization**

  All column names were renamed to snake_case format to improve code readability and consistency.

- **Step 4: Feature Engineering**
  - age_group column created by binning continuous customer age values into defined ranges.
  - purchase_frequency_days column derived from purchase frequency data.

- **Step 5: Redundancy Check**

  A consistency check confirmed that discount_applied and promo_code_used were redundant. The promo_code_used column was subsequently dropped.

- **Step 6: Database Integration**

  The cleaned DataFrame was exported from Python and loaded into MySQL database for structured SQL analysis.

4. Data Analysis using SQL (MySQL)

Structured queries executed in MySQL to extract answers to the following key business questions:

4.1 Revenue by Gender

Total revenue was compared between male and female customer segments.

gender	revenue
Male	157890
Female	75191

2 rows in set (0.01 sec)

4.2 High-Spending Discount Users

Customers who applied a discount but still recorded a purchase amount above the dataset average were identified.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67

1652	80
1654	93
1656	81
1659	66
1662	86
1667	64
1671	73
1673	73
1674	62
1676	90

839 rows in set (0.01 sec)

4.3 Top 5 Products by Review Rating

Products were ranked by their average review rating to surface highest-satisfaction items.

item_purchased	Average Product Rating
Gloves	3.8614285714285725
Sandals	3.8443750000000003
Boots	3.8187500000000005
Hat	3.8012987012987005
Skirt	3.784810126582278

5 rows in set (0.01 sec)

4.4 Shipping Type Comparison

Average purchase amounts were compared between Standard and Express shipping options.

shipping_type	ROUND(AVG(purchase_amount),2)
Express	60.48
Standard	58.46

2 rows in set (0.01 sec)

4.5 Subscribers vs. Non-Subscribers

Average spend per customer and total revenue were compared across subscription status groups.

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

2 rows in set (0.01 sec)

4.6 Discount-Dependent Products

The five products with the highest percentage of purchases involving a discount were identified.

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

5 rows in set (0.01 sec)

4.7 Customer Segmentation

Customers were classified into three segments — New, Returning, and Loyal — based on their historical purchase count.

customer_segment	no of customers
LOYAL	3116
RETURNING	701
NEW	83

3 rows in set (0.01 sec)

4.8 Repeat Buyers & Subscription Likelihood

Customers with more than 5 recorded purchases were analyzed to determine their correlation with subscription status.

subscription_status	repeat_buyers
Yes	958
No	2518

2 rows in set (0.01 sec)

4.9 Revenue Contribution by Age Group

Total revenue was aggregated by age group to identify the most valuable customer age segments.

age_group	total_revenue
Young	62143
Middle_Age	59197
Adult	55978
Senior	55763

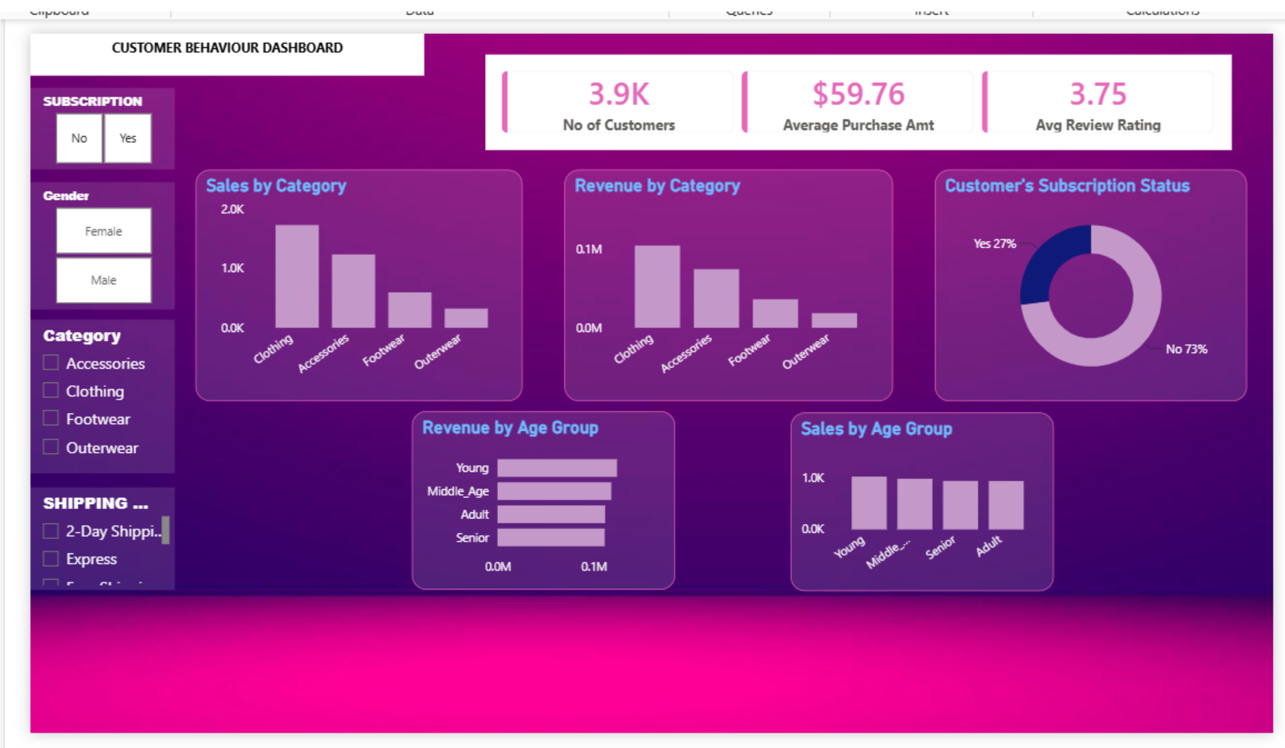
4 rows in set (0.01 sec)

5. Power BI Dashboard

Built an interactive dashboard in **POWER BI** to present insights visually.

The dashboard includes KPI cards, donut charts, bar charts, and slicer panels for dynamic exploration by Subscription Status, Gender, Category, and Shipping Type.

Dashboard Screenshot:



6. Business Recommendations

Based on the analysis, the following strategic recommendations are proposed:

- **Boost Subscriptions:**
  Promote exclusive benefits for subscribers.
- **Customer Loyalty**:
  Introduce rewards to systematically convert Returning customers (701) into the Loyal segment, which already represents 3,116 customers.
- **Margin Management:**
  Items like Hat, Sneakers, and Coat show ~50% discount dependency. Review whether discounting is driving incremental purchases or eroding margins on customers who would buy regardless.
- **Product Positioning:**
  Feature top-rated products (Gloves, Sandals, Boots) and best-sellers (Pants, Jewelry) prominently in marketing campaigns and homepage placements.
- **Targeted Marketing:**
  Young Adult customers generate the highest revenue ($62,143). Align digital ad spend toward this segment. Express shipping users show higher average spend ($60.48 vs $58.46) — upsell express shipping at checkout.